Design of a CMOS Two-Stage Miller Compensated OTA

Muyang Chen
University of Texas at Austin

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1 Introduction

The objective of this project is to design a CMOS two-stage Miller compensated operational transconductance amplifier (OTA) based on the TSMC 180nm process. This report serves as a practical engineering log, documenting my design flow, reasoning, and key insights throughout the design process. The design methodology references public textbooks and materials.

2 Target Specifications

The objective is to maximize the Unity-Gain Bandwidth (GBW) while strictly satisfying the following constraints operating on a 1.8V supply ($V_{DD} = 1.8$V, $V_{SS} = 0$V):

Parameter	**Target Specification**
Channel Length (L)	$\geq 0.8\,\mu$m
Channel Width (W)	$\geq 2\,\mu$m
Load Capacitance (C_L)	3 pF
Input Common-Mode ($V_{in,cm}$)	$(V_{DD} + V_{SS})/2$
Output Swing	$[0.1(V_{DD} - V_{SS}), 0.9(V_{DD} - V_{SS})]$
Static Power Consumption	≤ 2 mW
Open-Loop DC Gain (A_{v0})	≥ 80 dB
Unity-Gain Bandwidth (GBW)	Maximize
Phase Margin (PM)	$\geq 60°$
Slew Rate (SR)	$\geq 30\,\mathrm{V}/\mu$s
CMRR	≥ 60 dB
PSRR-	≥ 80 dB
Equivalent Input Noise	$\leq 300\,\mathrm{nV}/\sqrt{\mathrm{Hz}}$ @ 1kHz

3 Circuit Architecture & Analysis

The overall structure consists of four main blocks: the first-stage differential input amplifier, the second-stage common-source amplifier, the biasing network, and the phase compensation circuit.

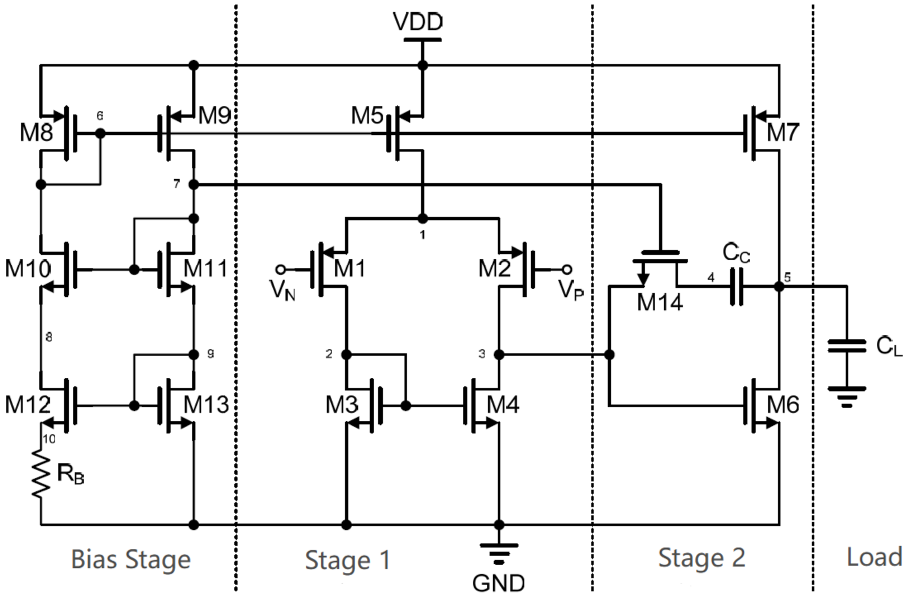

Figure 1: Overall Schematic of the Two-Stage Miller Compensated OTA

3.1 Gain

The total DC open-loop gain is the product of the gains from the first and second stages:

$$A_{v0} = g_{m2}(r_{o2} \parallel r_{o4}) \cdot g_{m6}(r_{o6} \parallel r_{o7}) \tag{1}$$

3.2 Frequency Response

The frequency response of this circuit is governed by three poles and two zeros:

a) **Dominant Pole (ω_d):**
Located at the output of the first stage. It is primarily formed by the first-stage output resistance and the Miller-multiplied compensation capacitor C_c:

$$\omega_d = \frac{1}{(r_{o2} \parallel r_{o4}) \cdot g_{m6}(r_{o6} \parallel r_{o7}) \cdot C_c} \tag{2}$$

b) **Second Dominant Pole (ω_{nd}):**
Located at the output of the second stage. Because C_c is relatively large, the gate and drain of M6 are effectively shorted at high frequencies, yielding:

$$\omega_{nd} = \frac{g_{m6}}{C_c}\left(\frac{1}{1 + C_{n1}/C_c}\right) \tag{3}$$

c) **Right-Half-Plane (RHP) Zero (ω_z):**
Introduced by the feedforward path through the Miller capacitor C_c. The zero occurs at the frequency where the output voltage drops to zero ($v_{out} = 0$). Under this condition, the current through the active load M7 is zero ($i_{d7} = 0$), meaning the transconductance current from M6 is entirely supplied by the feedforward current through C_c. By equating these currents:

$$\begin{aligned} v_{in} \cdot sC_c &= g_{m6} \cdot v_{in} \\ s = \frac{g_{m6}}{C_c} &\implies \omega_z = \frac{g_{m6}}{C_c} \end{aligned} \tag{4}$$

d) **Parasitic Zero-Pole Pair:**
Located at Node 2, introduced by the parasitic capacitance C_{n1}.

Assuming a single-pole approximated system, the Unity-Gain Bandwidth (GBW) simplifies to:

$$GBW = A_{v0} \cdot \omega_d = \frac{g_{m2}}{C_c} \tag{5}$$

3.3 Output Swing

The allowable output swing bounds are theoretically limited by the overdrive voltages of the output stage: $[V_{ov6}, V_{DD} - V_{ov7}]$.

4 Amplifying Stage Design (g_m/I_D Methodology)

From the theoretical analysis, key parameters such as PM, GBW, and A_{v0} are directly tied to the small-signal parameters of the transistors. My strategy is to first deduce these small-signal targets based on the specifications, and then leverage the g_m/I_D methodology to dimension the physical transistors.

4.1 Extracting Small-Signal Targets

a) **Phase Margin Budgeting:**
The zero-pole pair introduced by C_{n1} is closely spaced, so I roughly estimate it consumes about 5° of the phase margin budget. The overall frequency response can be described as:

$$H(s) = A_{v0} \frac{1 - s/\omega_z}{(1 + s/\omega_d)(1 + s/\omega_{nd})} \tag{6}$$

To achieve a PM $\geq 60°$, the total phase shift at GBW must satisfy $\angle H(GBW) \geq -120°$:

$$-\arctan\left(\frac{GBW}{\omega_z}\right) - \arctan\left(\frac{GBW}{\omega_d}\right) - \arctan\left(\frac{GBW}{\omega_{nd}}\right) - 5° \geq -120° \tag{7}$$

In a single-pole approximation ($GBW = A_{v0} \cdot \omega_d$), $\arctan(GBW/\omega_d) \to 90°$. Rearranging gives:

$$\arctan\left(\frac{GBW}{\omega_z}\right) + \arctan\left(\frac{GBW}{\omega_{nd}}\right) \leq 25° \tag{8}$$

b) **Pole/Zero Placement:**
Since ω_z sits at a higher frequency, I allocate approximately 8° of PM degradation to it. This leaves $\arctan(GBW/\omega_{nd}) \leq 17°$.

- *For ω_{nd}:* $\omega_{nd} \geq \frac{GBW}{\tan(17°)} \approx 3.27\,GBW$. For redundancy, I mandate $\omega_{nd} \geq 4 \cdot GBW$.
- *For ω_z:* $\omega_z \geq \frac{GBW}{\tan(8°)} \approx 7.12\,GBW$. For redundancy, I mandate $\omega_z \geq 8 \cdot GBW$.

c) **Calculating Parameters:**
Based on the derivation above, the following system of equations is formed:

$$\omega_z = \frac{g_{m6}}{C_c} = 8 \cdot 2\pi \cdot GBW \tag{9}$$

$$2\pi \cdot GBW = \frac{g_{m2}}{C_c} \tag{10}$$

$$\omega_{nd} = \frac{g_{m6}}{C_L}\left(\frac{1}{1 + C_{n1}/C_c}\right) = 4 \cdot 2\pi \cdot GBW \tag{11}$$

Notice that there are 5 unknown variables $(g_{m2}, g_{m6}, C_c, C_L, C_{n1})$ but only 3 equations. To close the system:

1) I first set a design target for GBW: $GBW = 75\,\text{MHz}$.
2) The parasitic capacitance C_{n1} at Node 2 is typically estimated within $0.1C_c < C_{n1} < 0.3C_c$. I take the worst-case scenario: $C_{n1} = 0.3C_c$.
3) To account for additional parasitic loading at the output node, I budget extra margin for C_L, setting $C_L = 4\,\text{pF}$ (compared to the 3 pF spec).

Solving this system yields the small-signal targets (with slight margins added):

$$g_{m2} = 0.825\,\text{mS}, \quad g_{m6} = 7.5\,\text{mS}, \quad C_c = 2\,\text{pF} \tag{12}$$

d) **Simulation Check:**
Plugging these parameters into an AC simulation yielded GBW = 62 MHz and PM = 67°. The small-signal targets are solid.

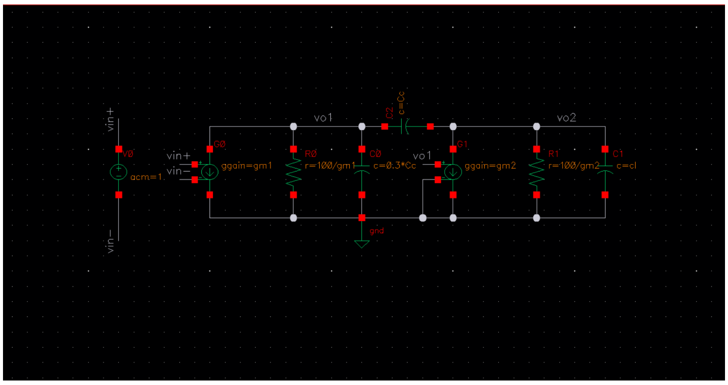

Figure 2: Small-Signal Model for the AC Simulation

4.2 Calculating Curve Constraints (g_m/I_D)

For each transistor, we have three core design variables: W, L, V_{gs}. I first derive constraints for $g_m r_o$, g_m/I_D, and g_m/C_{gg}, all of which are weakly dependent on W. By scanning these curves against L and V_{gs}, I can first lock in L and V_{gs}. Once they are determined, I then scan g_m and I_D against W to select an appropriate channel width W that hits the target transconductance and bias current (using a default multiplier of 4 during scans).

a) **Intrinsic Gain ($g_m r_o$):**
To hit $A_{v0} \geq 80\,\text{dB}$, $g_m r_o > 40\,\text{dB}$ per stage is required. Adding margin, I target $g_m r_o > 46\,\text{dB} \approx 200$.

b) **Current Efficiency (g_m/I_D):**
Bounded by the 2mW power budget, $I_{D,spec} \leq 1.1\,\text{mA}$.

$$\frac{2g_{m2} + g_{m6}}{2I_{d2} + I_{d6}} \geq \frac{2g_{m2} + g_{m6}}{0.75 \cdot I_{D,spec}} = 11.1\,\text{V}^{-1} \quad \text{(0.75 is a margin factor)} \tag{13}$$

Thus, $g_{m2}/I_{d2}, g_{m6}/I_{d6} \geq 11.1\,\text{V}^{-1}$.

Furthermore, to meet the lower output swing bound of $0.1(V_{DD} - V_{SS})$, $V_{ov6} \leq 0.18\text{V}$ must hold. Since $g_{m6} = 2I_{d6}/V_{ov6}$, this also requires $g_{m6}/I_{d6} \geq 11.1\,\text{V}^{-1}$, perfectly aligning with the current constraint.

c) **Transit Frequency (g_m/C_{gg}):**
To keep the parasitic pole high enough, I assume $C_{gg2} + C_{gg6} \leq 0.75C_{n1}$.

$$\frac{g_{m2}}{2\pi f_{T2}} + \frac{g_{m6}}{2\pi f_{T6}} \leq 0.45\,\text{pF} \implies f_T \geq 3\,\text{GHz} \tag{14}$$

However, practically, this constraint is too harsh for the first-stage PMOS input pair. Since only the drain capacitance of the input pair contributes to C_{n1} (usually $< 0.5C_{gg}$), I relax the f_T requirement for the first stage to 1.2GHz. Thus: $f_{T2} \geq 1.2\,\text{GHz}$, $f_{T6} \geq 3\,\text{GHz}$.

d) **Current (I_d):**
Provisionally setting the bias current source to $5\,\mu$A, I_{d2} will be $n \times 2.5\mu$A, and I_{d6} will be $n \times 5\mu$A.

4.3 Translating to Transistor Sizing

Scanning M2 and M6 across these constraint curves, I selected appropriate V_{gs} and L:

$$V_{gs2} = 586\,\text{mV},\ L_2 = 1\,\mu\text{m};\quad V_{gs6} = 586\,\text{mV},\ L_6 = 1\,\mu\text{m}$$

Scanning g_m and I_D against W (with multiplier $m = 4$), I chose:

$$W_2 = 31.3\,\mu\text{m},\ I_{d2} = 90\,\mu\text{A},\ g_{m2} = 1.03\,\text{mS}$$

$$W_6 = 44.34\,\mu\text{m},\ I_{d6} = 650\,\mu\text{A},\ g_{m6} = 8\,\text{mS}$$

Why does the second stage draw so much more current than the first?
To meet the Phase Margin requirements, the transconductance of the second stage amplifying transistor (g_{m6}) must be significantly larger than the first. Since $g_m = \mu C_{ox}(W/L)V_{ov}$, boosting g_m while keeping V_{ov} constant means we have to drastically widen the transistor (W). Because $I_D \propto (W/L)V_{ov}^2$, the current also scales linearly with W. Therefore, to achieve that necessary transconductance, the second stage inherently consumes much more current.

4.4 Simulation Verification

To verify the core amplifying path, I ran an STB simulation on a hybrid testbench. In this setup, only the designed amplifying transistors (M1, M2, and M6) were used as real devices, while all remaining components were replaced with ideal elements to isolate the amplifying performance:

a) The current sources (M5 and M7) were replaced by ideal DC current sources I_0 ($180\,\mu$A) and I_1 ($650\,\mu$A).

b) The active current mirror load (M3 and M4) was completely replaced by a combination of an ideal DC voltage source V_0 and a Current-Controlled Current Source (CCCS) F_0.

c) To ensure proper DC matching, V_0 was set to 586 mV (matching V_{gs6} to clamp the first-stage output node to the exact bias voltage required by M6), while F_0 mirrored the branch current flowing through V_0.

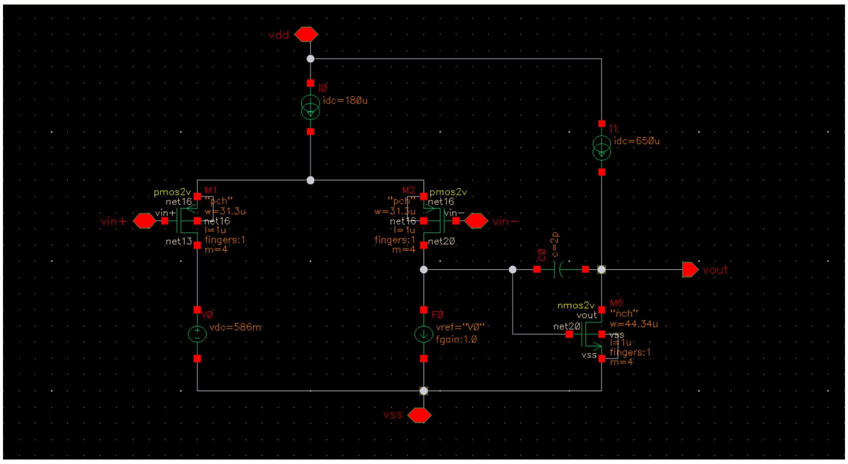

Figure 3: STB Simulation Setup for M2/M6 Verification

The simulation yielded **GBW = 74.2 MHz, PM = 60.7°, Av = 94 dB**. The Phase Margin was a bit too close for comfort.

a) First, I introduced an ideal nulling resistor (tentatively $1/g_{m6} \approx 125\,\Omega$), pushing PM to 66.8°.

b) Next, I scaled down the W and L of M1, M2, and M6 by a factor of 0.8. Since these parasitic capacitances load Node 2 and the output, scaling them down (without altering the DC operating point) pushed the PM up to 69°, at the minor cost of slightly reduced gain.

Amplifying transistor parameters obtained:

$$L_1 = L_2 = 0.8\,\mu\text{m}, \quad W_1 = W_2 = 25.04\,\mu\text{m}$$

$$L_6 = 0.8\,\mu\text{m}, \quad W_6 = 35.47\,\mu\text{m}$$

5 Load Transistors Design

For the load transistors, the goals are straightforward: high output impedance and low parasitic capacitance.

5.1 Curve Constraints

a) **Output Conductance (g_{ds}/I_D):**
$g_{ds} = 1/r_o$ is used. (Note: $r_o/I_D \propto 1/W^2$ is strongly dependent on W, violating the methodology). To limit attenuation to 6 dB per stage, the load's r_o must be at least equal to the amplifying transistor's r_o. From previous DC simulations ($g_{ds2} = 5.27\,\mu\text{S}$ and $g_{ds6} = 31.69\,\mu\text{S}$), I constrain:

$$g_{ds4}/I_{d4} \leq 5.27\mu\text{S}/90\mu\text{A} = 0.0586\,\text{V}^{-1}$$

$$g_{ds7}/I_{d7} \leq 31.69\mu\text{S}/650\mu\text{A} = 0.0488\,\text{V}^{-1}$$

b) **Parasitic Capacitance (C_{gg}/I_D):**
For M3 (loading Node 2), I constrain $C_{gg4} \leq C_{n1}/3 = 0.1C_c$, giving $C_{gg4}/I_{d4} \leq 2.22\,\text{nF/A}$. For M7 (loading the output), $C_{dd7} \leq 0.2C_L$, yielding $C_{dd7}/I_{d7} \leq 923\,\text{pF/A}$.

5.2 Sizing the Load Transistors

Scanning M4 (to match voltages at Nodes 2 and 3 and minimize mismatch), V_{gs4} must equal $V_{gs6} = 586\,\text{mV}$. I chose $L = 1\,\mu\text{m}$ and $W = 6.2\,\mu\text{m}$.

For M7, I_{d7} is locked at $650\,\mu\text{A}$ (130 times the $5\,\mu\text{A}$ bias source), so the multiplier is set to 130 for scanning.

The Output Swing Bottleneck

Observing the curves, both metrics improve as V_{sg} rises. So, should we just blindly crank up V_{sg}? Absolutely not. We must respect the output swing requirement of $0.9(V_{DD} - V_{SS}) = 1.62\text{V}$. To keep M7 saturated:

$$V_{sd7} > V_{sg7} - |V_{th}| \implies V_{DD} - V_{out} > V_{sg7} - |V_{th}|$$

$$V_{DD} - V_{ov7} > 1.62\text{V} \implies V_{ov7} < 0.18\text{V}$$

Since $V_{ov7} = 2I_{d7}/g_{m7} < 0.18\text{V}$, $g_{m7}/I_{d7} > 11.1\,\text{V}^{-1}$ is required. Balancing this, I picked $V_{gs} = 590\,\text{mV}$, $L = 0.8\,\mu\text{m}$, $W = 5.3\,\mu\text{m}$.

Due to the current mirror setup, M5 matches M7 with a multiplier of $180/5 = 36$. Load transistor parameters obtained:

$$L_3 = L_4 = 1\,\mu\text{m}, \quad W_3 = W_4 = 6.2\,\mu\text{m}$$
$$L_5 = L_7 = 0.8\,\mu\text{m}, \quad W_5 = W_7 = 5.3\,\mu\text{m}$$

6 Simulation & Iterative Optimization

I ran an STB simulation on the complete first and second stages using a $5\,\mu\text{A}$ ideal current source.

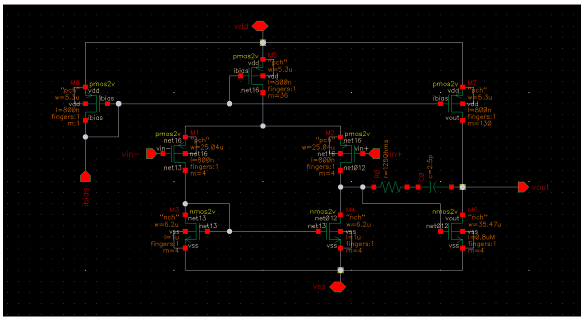

Figure 4: Core Circuit Schematic

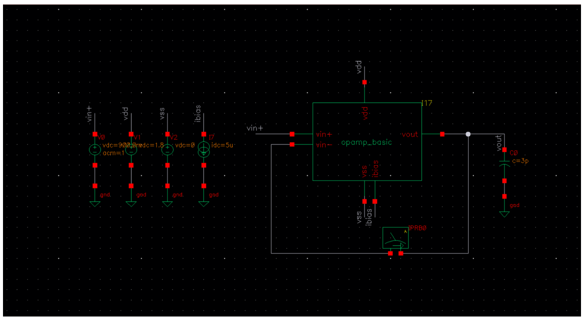

Figure 5: Testbench Setup

Initial Result: Gain = 81.3 dB, GBW = 70.74 MHz, I_{dc} = 833 µA, PM = 63.9°, Noise = 74.4 nV/$\sqrt{\text{Hz}}$ @ 1kHz.

The Gain and Phase Margin (PM) were uncomfortably close to the specification limits. Earlier, shrinking M1 and M2 bought some PM by reducing parasitic capacitances, but it noticeably degraded the DC Gain (A_{v0}). To fix this, I revisit the fundamental intrinsic gain equation:

$$A_v = g_m r_o = \frac{g_m}{\lambda I_D} = \frac{1}{\lambda}\sqrt{\frac{2\mu C_{ox}(W/L)}{I_D}} \propto \sqrt{\frac{L}{I_D}} \tag{15}$$

Since DC Gain is directly proportional to $\sqrt{L/I_D}$, I can boost the gain by decreasing I_D and increasing L. To recover the lost PM caused by these adjustments, I can scale up the Miller compensation capacitor C_c, sacrificing GBW as an acceptable trade-off.

Modify the First Stage or the Second Stage?

- **Option A: Modifying the First Stage**
  - *Pros:* Lowering the first-stage bias current (I_{d5}) decreases g_{m1}. While this trades off GBW, it naturally improves Phase Margin (PM) and boosts Gain ($A_{v0} \propto 1/\sqrt{I_D}$). Additionally, because SR $\approx (I_{d7} - I_{d5})/C_L$, reducing I_{d5} directly improves the Slew Rate.
  - *Cons:* GBW takes a direct hit, making it harder to maintain high-speed performance.
- **Option B: Modifying the Second Stage**
  - *Pros:* Allows scaling output impedance R_{out} and overall Gain without sacrificing first-stage transconductance (g_{m1}).
  - *Cons:* Reducing the second-stage current or increasing its channel length (L_7) reduces g_{m6} and increases output parasitic capacitance, which lowers the non-dominant pole ω_{nd} and degrades PM. Furthermore, a larger L_7 increases V_{ov7}, pinching the output voltage swing.

Therefore, I chose to tune the second stage to recover the required DC Gain while preserving first-stage transconductance for bandwidth, offsetting PM degradation by scaling C_c.

The Fix:

1) **Fix Second-Stage Output Impedance:** DC simulation revealed that $r_{o7} = 26.8\,\text{k}\Omega$ while $r_{o6} = 31.5\,\text{k}\Omega$. Clearly, M7's lower output resistance was dragging down the total output impedance. I re-sized M7 by increasing its length to $L_7 = 1\,\mu\text{m}$ and $W_7 = 6.57\,\mu\text{m}$ to boost its intrinsic impedance.

2) **Reduce First-Stage Currents & Impedance Matching:** I dropped I_{d5} from $180\,\mu\text{A}$ down to $123\,\mu\text{A}$. This current drop reduced V_{gs4}. To restore DC matching so that V_{gs4} once again equals V_{gs6}, I lengthened L_4. Lengthening L_4 also bumped up r_{o4}, so I proportionally scaled up W_2 and L_2 to ensure r_{o2} increased in tandem, maintaining $r_{o1,2} \approx r_{o3,4}$.

3) **Adjust Miller Capacitance:** I scaled up the Miller compensation capacitor C_c from 2 pF to 2.8 pF to increase Phase Margin (PM).

4) **Current Mirror Fine-Tuning:** To compensate for residual mismatch errors and guarantee that I_{d7} strictly holds at $650\,\mu\text{A}$, I tweaked M7's multiplier (m) to 128.

Optimized Result: **Gain = 86.4 dB, GBW = 40.7 MHz, PM = 66.9°**. A much more robust design point.

Now that I have obtained the first and second stage transistor parameters, the dimensions are summarized in Table 1:

Table 1: Summary of Final Transistor Sizing (Amplifying & Active Load Stages)

Transistor	**Type**	**Width (W) [μm]**	**Length (L) [μm]**	**Multiplier (m)**
M1, M2	PMOS	50.00	1.60	4
M3, M4	NMOS	6.20	1.70	4
M5	PMOS	6.57	1.00	25
M6	NMOS	35.47	0.80	4
M7	PMOS	6.57	1.00	128

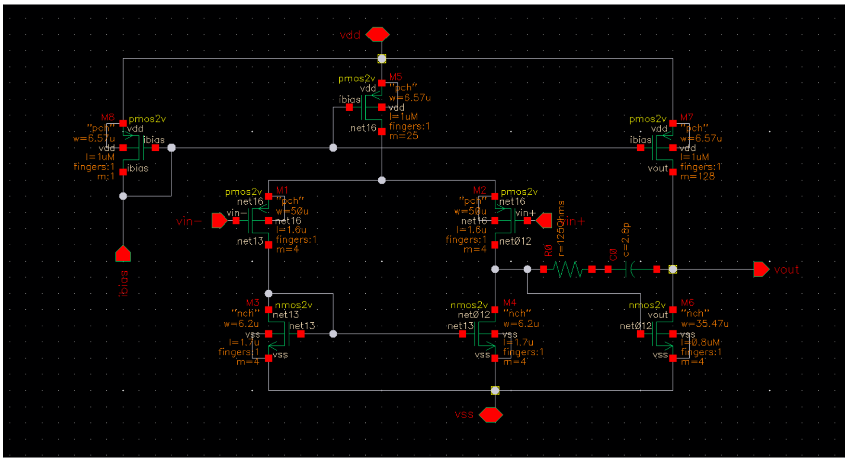

Figure 6: Optimized Core Schematic

7 Biasing Circuit Design

As dictated by the system requirements, the biasing circuit must generate a stable reference current of $I_b = 5\,\mu\text{A}$.

7.1 Current Reference Derivation

Since M8 and M9 form a current mirror, $I_{d12} = I_{d13}$ holds. Using the square-law characteristic:

$$\left(\frac{W}{L}\right)_{12} V_{ov12}^2 = \left(\frac{W}{L}\right)_{13} V_{ov13}^2 \tag{16}$$

By designing $W_{12} = 4 \cdot W_{13}$, it follows that $V_{ov13} = 2V_{ov12}$.

$$\begin{aligned} V_{gs13} &= R_b I_b + V_{gs12} \\ V_{ov13} &= R_b I_b + V_{ov12} \implies 2V_{ov12} = R_b I_b + V_{ov12} \implies V_{ov12} = I_b R_b \end{aligned} \tag{17}$$

Substituting V_{ov12} back into the expression for $I_b = I_{d12}$:

$$I_b = \frac{1}{2}\mu_n C_{ox} \left(\frac{W}{L}\right)_{12} V_{ov12}^2 = \frac{1}{2}\mu_n C_{ox} \left(\frac{W}{L}\right)_{12} I_b^2 R_b^2 \implies I_b = \frac{2}{\mu_n C_{ox} (W/L)_{12} R_b^2} \tag{18}$$

Notice that I_b relies solely on M12's dimensions and the resistance R_b, completely independent of the power supply voltage (V_{DD}). R_b is implemented off-chip, modeled here by an ideal resistor.

7.2 Transistor Sizing & Biasing Voltage Constraints

M8 and M9 share the same channel length and width as M5 and M7, with a multiplier of $m = 1$. M10–M11 and M12–M13 form a cascode structure to isolate supply fluctuations and suppress channel-length modulation. Therefore, L_{10} and L_{11} must be relatively large to maximize output resistance, while L_{12} and L_{13} are also made large to minimize channel-length modulation.

When sizing M10–M13, two key voltage headroom constraints must be satisfied:

1) **Biasing M14 efficiently:**
The drain voltage of M11 provides the gate bias for M14 ($V_{d11} = V_{gs11} + V_{gs13}$). Since M14's linear-region resistance is inversely proportional to $V_{ov14}(W/L)_{14}$, a higher gate bias voltage (V_{d11}) reduces the required area of M14, minimizing parasitic capacitance. This implies $(W/L)_{13}$ and $(W/L)_{14}$ should be kept small.

2) **Matching M8/M9 drain voltages:**
To keep $V_{d8} \approx V_{d9}$ and minimize M9's channel-length modulation, V_{d11} should sit at a relatively high potential:

$$V_{d11} \approx V_{DD} - V_{sg8} = V_{gs11} + V_{gs13} \tag{19}$$

Given $V_{sg8} = V_{sg7} = 590\,\text{mV}$, I obtain $V_{gs11} \approx V_{gs13} \approx 605\,\text{mV}$.

Setting $W_{11} = 2\,\mu\text{m}$ with $m = 2$, $V_{gs} = 605\,\text{mV}$, and $I_d = 5\,\mu\text{A}$, I scanned L and locked $L_{11} = 4.53\,\mu\text{m}$.

Using these parameters to construct the complete biasing circuit and scanning R_b, I obtained $R_b = 21.9\,\text{k}\Omega$, drawing the target $5\,\mu\text{A}$ bias current as desired.

DC simulation confirms that the operating points and voltages on both sides of the bias network are perfectly symmetrical.

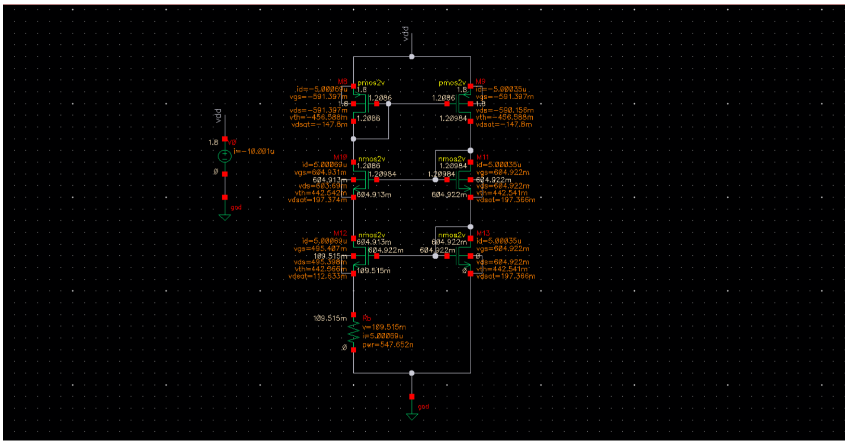

Figure 7: Biasing Circuitry with Annotated DC Operating Points

Thus, the final sizing for the cascode network is:

$$W_{10} = W_{11} = W_{12} = W_{13} = 2\,\mu\text{m}$$

$$L_{10} = L_{11} = L_{12} = L_{13} = 4.53\,\mu\text{m}$$

$$\text{Multipliers: } m_{10} = m_{11} = m_{13} = 2,\ m_{12} = 8$$

8 The Nulling Resistor (R_z)

8.1 Analysis of R_z

Revisiting the zero analysis in Section 3.2, I introduce a nulling resistor R_z in series with C_c. The modified zero equation becomes:

$$\omega_z = \frac{1}{C_c\left(\frac{1}{g_{m6}} - R_z\right)} \tag{20}$$

By tuning R_z, I can control the location of the zero. In general, there are three distinct strategies to manage this zero:

1) **Cancel the non-dominant pole (ω_{nd}):** By placing the zero in the LHP directly on top of ω_{nd}, one can eliminate the second pole. This requires $R_z = \frac{1}{g_{m6}}\left(1 + \frac{C_L}{C_c}\right)$. However, due to PVT variations and load capacitance shifts in practical circuits, exact cancellation is nearly impossible, easily creating pole-zero doublets that severely degrade transient response. Thus, this method is rejected.

2) **Push the zero to infinity:** Setting $R_z = 1/g_{m6}$ cancels the feedforward current, moving $\omega_z \to \infty$. Although transistor ratio-matching can theoretically render this PVT-independent, short-channel devices deviate significantly from square-law behaviors, introducing calculation errors. Thus, this method is also rejected.

3) **Move the zero into the Left-Half-Plane (LHP) (Chosen):** Once moved into the LHP, the zero starts contributing phase lead instead of phase lag, boosting the Phase Margin (PM).

> **Negative impact of Low-Frequency Zeros**
>
> One might be tempted to select a hugely oversized R_z to pull the LHP zero down to a very low frequency, which drastically inflates the frequency-domain Phase Margin by tens of degrees. However, placing a zero at low frequencies creates a closely spaced **pole-zero doublet**. This doublet degrades settling time and severe ringing in the transient step response and invalidates the single-pole system approximation. Therefore, the zero must be placed in the LHP at a frequency above GBW to provide phase lead without corrupting transient response.

I chose to place the LHP zero at $3 \cdot GBW$. Setting $\omega_z = -3\frac{g_{m1}}{C_c}$, I solve for R_z:

$$-3\frac{g_{m1}}{C_c} = \frac{1}{C_c\left(\frac{1}{g_{m6}} - R_z\right)} \implies R_z = \frac{1}{g_{m6}} + \frac{1}{3g_{m1}} \approx 500\,\Omega \tag{21}$$

8.2 Transistor Implementation of R_z

R_z is implemented using M14 operating in the triode region. Based on DC simulation, the gate-source voltage of M14 is $V_{gs14} = 1.21\,\text{V} - 587.1\,\text{mV} = 623\,\text{mV}$. With $V_{gs14} = 623\,\text{mV}$, $V_{ds14} = 0\,\text{V}$, and target resistance $1/g_{ds14} = 500\,\Omega$, I scanned the dimensions and locked:

$$L_{14} = 1.12\,\mu\text{m}, \quad W_{14} = 10\,\mu\text{m} \tag{22}$$

8.3 Iterative Optimization via C_c

Replacing the ideal resistor with M14 in the circuit, the simulation yielded:

$$\text{Gain} = 86.4\,\text{dB}, \quad \text{PM} = 80^\circ, \quad \text{GBW} = 43\,\text{MHz}, \quad \text{Noise} = 42.5\,\text{nV}/\sqrt{\text{Hz}}@1\text{kHz}, \quad I_{dc} = 783\,\mu\text{A}$$

The nulling resistor provided a massive 14° boost in Phase Margin! Capitalizing on this extra phase headroom, I reduced C_c from 2.8 pF down to 1.8 pF to reclaim bandwidth.

Final Core Performance: **Gain = 87 dB, PM = 66.4°, GBW = 61 MHz, Noise = 42.5 nV/$\sqrt{\text{Hz}}$ @ 1kHz, I_{dc} = 783 µA**. All core parameters satisfy the target specs.

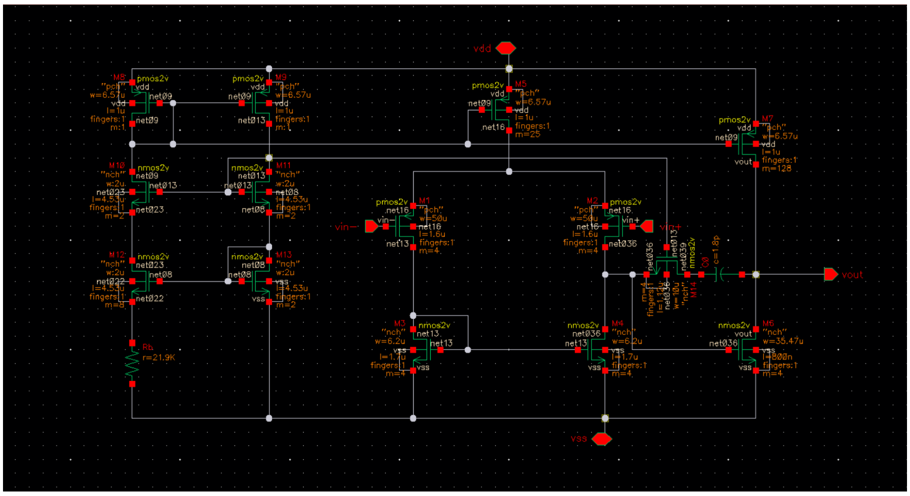

Figure 8: Final Complete OTA Schematic

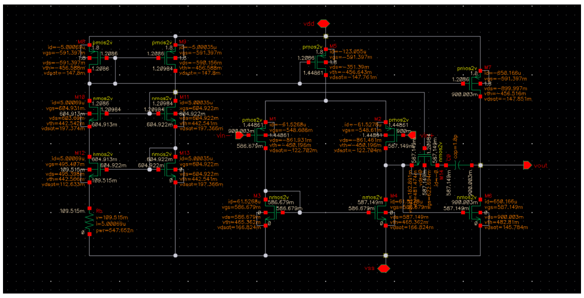

Figure 9: DC Operating Points Annotation

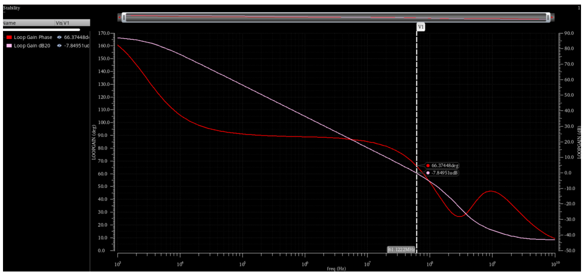

Figure 10: Open-Loop Bode Plot Response

9 Final Verification (SR, CMRR, PSRR & PVT)

Having verified the DC gain, phase margin, GBW, noise, and power consumption, I next evaluate the remaining specifications to ensure full compliance and make design iterations as needed.

9.1 Slew Rate & Common-Mode Input Range Analysis

Slew Rate (SR) measures the amplifier's large-signal transient speed. I configure the OTA as a unity-gain follower, apply a step input, and measure the rate of change of the output voltage. To determine a valid step amplitude, I first analyze the input common-mode range.

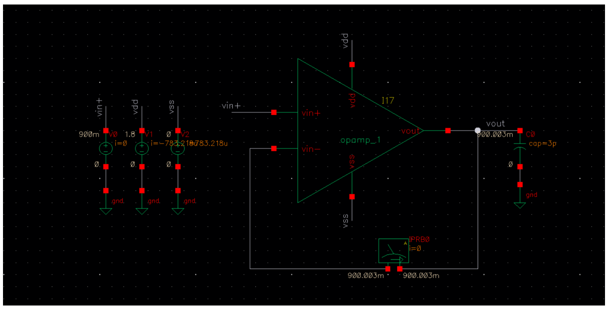

Figure 11: Slew Rate & Transient Response Testbench

9.1.1 Common-Mode Input Bounds Analysis

1) **Low-Level Input Limit:**
When the input signal drops to a low level, the voltage at Node 2 rises, which risks pulling M6 into the triode region. To keep M6 saturated:

$$V_{in} = V_{out} > V_{gs6} - V_{th6} \tag{23}$$

Plugging in the DC operating values from the $V_{cm} = 0.9\,\text{V}$ bias point yields $V_{in} > 0.1\,\text{V}$.

2) **High-Level Input Limit:**
To maintain the tail current source M5 in saturation ($V_{g5} > V_{d5} - |V_{th5}|$):

$$V_{DD} - V_{sg8} > V_{cm} + V_{sg2} - |V_{th5}| \implies V_{cm} < V_{DD} - V_{sg8} - V_{sg2} + |V_{th5}| \tag{24}$$

Plugging in numbers gives $V_{cm} < 1.116\,\text{V}$.

However, for large-signal slewing, the true bottleneck for Slew Rate is the charging current supplied by M7. As long as V_{cm} does not go high enough to drag M7 into its triode region, the circuit can slew properly:

$$V_{sd7} > V_{sg7} - |V_{th7}| \implies V_{DD} - V_{cm} > V_{sg7} - |V_{th7}| \implies V_{cm} < V_{DD} - V_{sg7} + |V_{th7}| \tag{25}$$

Plugging in the DC bias numbers yields a relaxed upper threshold of $V_{cm} < 1.665\,\text{V}$.

Simulations validate this insight: if the high-level input exceeds $1.6\,\text{V}$, M6's gate voltage is pulled excessively low, driving M7 into the triode region. Consequently, M7's charging current collapses below $650\,\mu\text{A}$, causing the output voltage to suffer from an extremely slow rising tail (visible in Fig. 12).

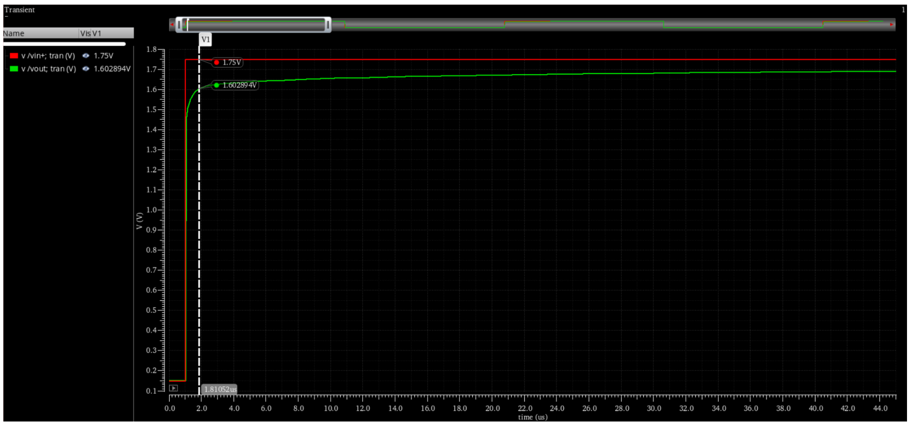

Figure 12: Output voltage's slow response when $V_{cm} > 1.6$V

9.1.2 Slew Rate Measurement

To stay safely within the non-distorted slewing boundary, I applied a pulse input stepping from 0.15 V to 1.4 V. Using the Cadence `slewrate` calculator function across the 10%-to-90% output region, I obtain:

$$\mathbf{SR+} = +\mathbf{33.43}\,\mathrm{V}/\mu\mathrm{s}, \quad \mathbf{SR-} = -\mathbf{78.57}\,\mathrm{V}/\mu\mathrm{s} \tag{26}$$

Both positive and negative Slew Rates exceed the $\geq 30\,\mathrm{V}/\mu\mathrm{s}$ specification.

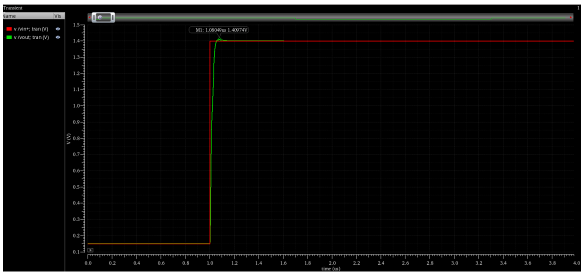

Figure 13: Positive Slew Rate (SR+) Transient Waveform

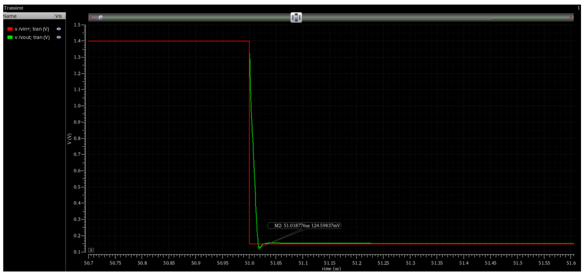

Figure 14: Negative Slew Rate (SR-) Transient Waveform

9.2 Common-Mode Rejection Ratio (CMRR)

To accurately simulate the Common-Mode Rejection Ratio (CMRR), the testbench must satisfy two conditions simultaneously:

1) **DC Operating Point:** The circuit must operate in a unity-gain negative feedback loop to stabilize the DC bias points.

2) **AC Common-Mode Signal:** The differential inputs (V_{in+} and V_{in-}) must be shorted together for small-signal AC signals.

To implement this idea, a 1 GF capacitor is connected in parallel between the two input terminals, and a 1 GH inductor is inserted in the feedback loop (as shown in Fig. 15).

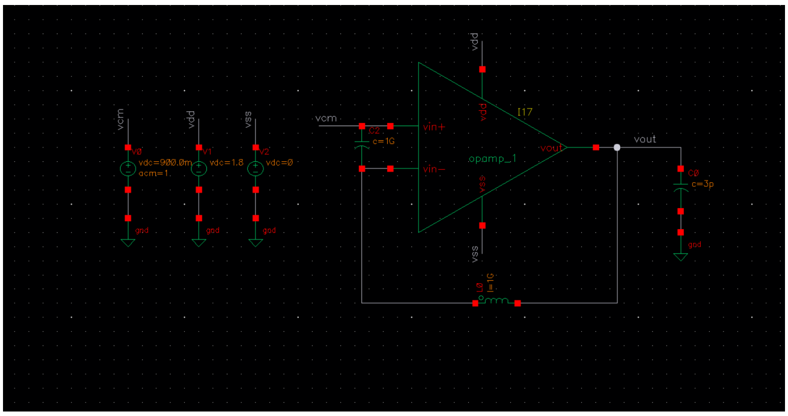

Figure 15: CMRR Testbench using 1 GH Inductor and 1 GF Capacitor

Running an AC simulation yields a common-mode gain of $A_{cm} = -0.56\,\mathrm{dB}$ (as shown in Fig. 16). With a differential gain of $A_{dm} = 87\,\mathrm{dB}$, the resulting CMRR is:

$$\mathrm{CMRR} = A_{dm} - A_{cm} = 87\,\mathrm{dB} - (-0.56\,\mathrm{dB}) = 87.56\,\mathrm{dB} \tag{27}$$

This comfortably passes the target specification of $\mathrm{CMRR} \geq 60\,\mathrm{dB}$.

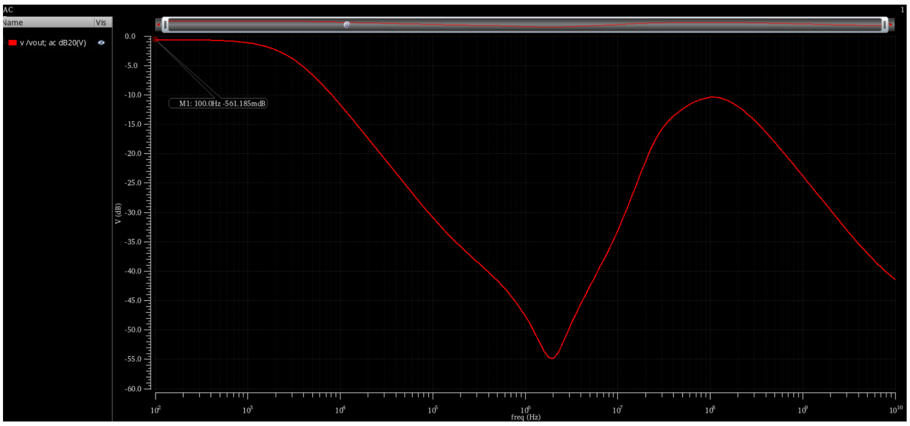

Figure 16: Common-Mode Gain (A_{cm}) Bode Plot Response

9.3 PSRR-

Setting the AC input to the V_{SS} rail, the negative supply gain measured 0.56 dB. Thus, PSRR- $= A_{dm} - 0.56\,\text{dB} = 86.44\,\text{dB}$.

This passes the target specification of PSRR- $\geq 80\,\text{dB}$.

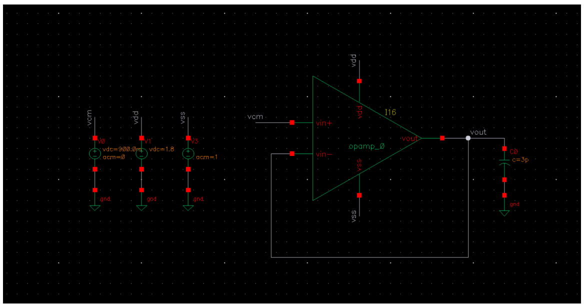

Figure 17: PSRR- Simulation Testbench

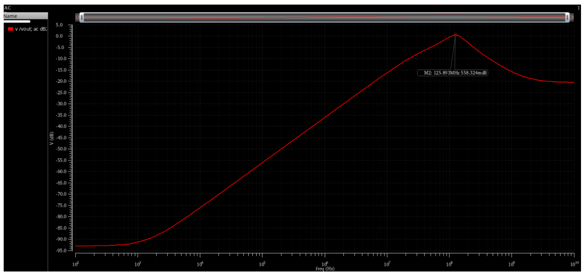

Figure 18: PSRR- AC Sweep

9.4 PVT Variations

To ensure the OTA consistently meets performance constraints across all operating conditions, I conducted a comprehensive 27-corner PVT sweep across key parameters: DC Gain, GBW, Phase Margin, Input Noise, and Power Dissipation.

The simulation matrix combines:

- **Process Corners:** tt, ff, ss
- **Supply Voltages (V_{DD}):** 1.6 V, 1.8 V, 2.0 V
- **Temperatures:** $-40°$C, 27°C, 85°C

Across all 27 PVT combinations, the circuit maintained robust performance bounds:

- **Open-Loop Gain (A_{v0}):** 81.61 dB $\sim$ 88.05 dB (Spec $\geq$ 80 dB)
- **Phase Margin (PM):** 63.36° $\sim$ 71.20° (Spec $\geq$ 60°)
- **Unity-Gain Bandwidth (GBW):** 51.28 MHz $\sim$ 65.45 MHz
- **Equivalent Input Noise (@ 1kHz):** $38.25\,\text{nV}/\sqrt{\text{Hz}} \sim 44.79\,\text{nV}/\sqrt{\text{Hz}}$ (Spec $\leq 300\,\text{nV}/\sqrt{\text{Hz}}$)

- **DC Current Consumption (I_{dc}):** $546.9\,\mu\text{A} \sim 1.041\,\text{mA}$

Output	Spec	Weight	Pass/Fail	Min	Max
gain_dc				81.61	88.05
Phase Margin				63.36	71.2
GBW				51.28M	65.45M
noise_1k				38.25n	44.79n
Idc				546.9u	1.041m
/V2/MINUS					

Figure 19: Summary Table of PVT Simulation Extrema

Conclusion: Even under severe PVT variations, the OTA satisfies all key specifications, demonstrating excellent design margin and reliability.

References

[1] B. Razavi, *Design of Analog CMOS Integrated Circuits*, 2nd ed. New York, NY, USA: McGraw-Hill Education, 2017.

[2] R. Yin, *Two-Stage Miller Compensated Operational Amplifier Design Tutorial* (in Chinese), Fudan University, Oct. 2007.

[3] IC Design Website, https://icdesign.com (accessed Jul. 2026).